

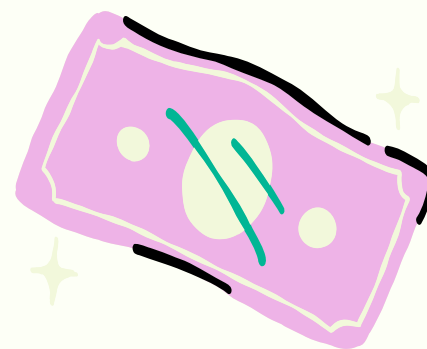
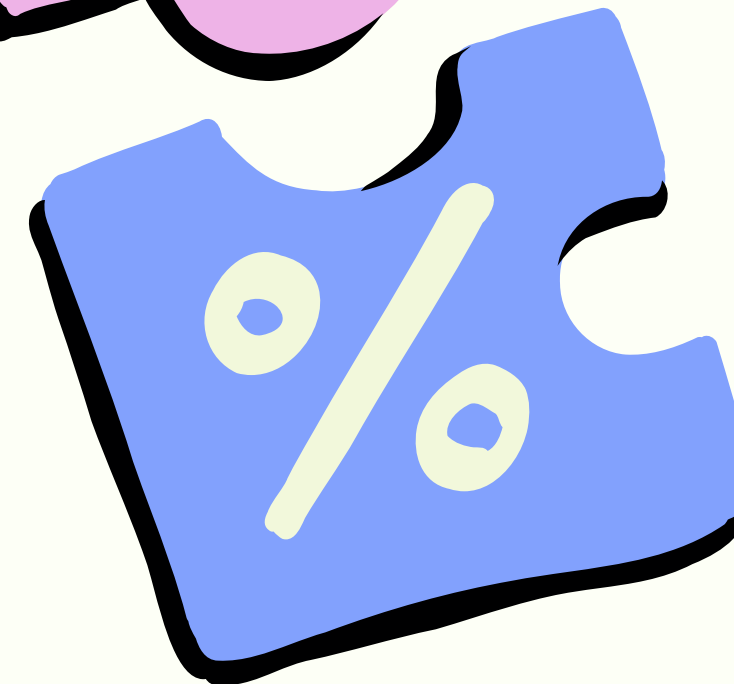
BUSINESS INTELLIGENCE

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1. INTRODUCTION:

This project focuses on analyzing automotive data using Microsoft Power BI. The main objective is to compare car brands, analyze prices, and identify the factors that affect vehicle value. The data used in

.this project was collected from Kaggle

The final outcome is an interactive dashboard that provides clear insights to support better decision-

.making in the automotive industry





2. PROBLEM DESCRIPTION:

In the automotive market, understanding car prices can be difficult because many factors influence their value. Many users lack simple tools to analyze car data and identify important patterns. Without clear insights into factors such as brand, model, year, and mileage, it becomes difficult to make informed decisions. In addition, the data may contain missing values and inconsistencies, which can reduce the accuracy of the analysis. This creates a need for a clear and interactive way to explore the data and support better decision-making.



3. OVERVIEW OF SOLUTION:

To address the challenges in understanding and analyzing car prices, this project uses Power BI to develop a clear and interactive dashboard. By presenting key factors such as brand, model, year, and mileage in visual form, users can easily explore the data and recognize patterns that affect car value. These visual insights improve the understanding of price variations and support users in making more accurate and informed decisions in the automotive market.



4. DESCRIPTION OF DATA:

Our data was collected from Kaggle. The first table contains car prices and serves as the main table, while the second table includes car specifications.

The main table consists of 558,837 records and 17 features. Examples of these features include the year, condition, selling price, and additional information such as state, location, type, and others [2].

The second table contains 1,500 records and 13 features, including model, seating capacity, rating, max power, fuel tank capacity, and other related features [3].



5. DATA PREPARATION :

Loaded both tables and checked data shape, data types, and missing values.

Removed unnecessary columns and handled missing values using row deletion, mode, and median replacement methods.

Standardized text columns by formatting values and removing extra spaces.

Converted numerical columns to the correct data types for accurate analysis.

Detected and removed outliers using the IQR method to improve data quality. Saved the cleaned datasets for further analysis and visualization.



6. VISUAL MODEL:

The data model was structured into three tables connected by a unique Car_Key to ensure clean relationships, eliminate redundancy, and support accurate analysis in Power BI.

car_prices_BI: Fact table containing all market and listing data (price, condition, color, odometer, seller, state). It is the primary source for pricing trends and KPIs.[2]

car_Specifications_BI: Dimension table with technical attributes (horsepower, RPM, torque, fuel tank, seating, rating). It supports performance comparisons and links specifications to pricing outcomes.[3]

car_prices_unique: Bridge table containing the unique Car_Key to resolve duplicates and convert Many-to-Many relationships into consistent One-to-Many filtering across visuals.

This three-table structure creates an organized, scalable, and analytically reliable model that enhances filtering accuracy and strengthens insights connecting car performance to market pricing.



7. DASHBOARD DESIGN OVERVIEW:

7.1 Dashboard Structure and Key Visual Elements

The dashboard is a single interactive page containing the key visuals for analyzing car performance, pricing, and distribution. All visuals update dynamically through slicers.

Total Cars: KPI showing dataset size

Performance KPIs: Highest horsepower (400 BHP) and max RPM (7K)

Average Rating by Make: Compares brand ratings

Average Horsepower by Make: Shows performance differences

Car Distribution by State: Map of regional availability

Average Selling Price by Make: Highlights pricing variations

Total Cars by Make: Donut chart showing brand share

Average Mileage & Max Price: Combined chart linking mileage with value



7. DASHBOARD DESIGN OVERVIEW:

7.2 Design and Aesthetic Choices

- Color Theme: A light background with soft blue and gray tones creates a clean, professional, and readable interface.
- Interactivity: Slicers allow users to explore data by car make and body type for more focused comparisons.
- Visual Consistency: Unified fonts, labels, and chart styles ensure clarity and a coherent analytical flow.
- Ease of Use: A simple, accessible layout enables all users to navigate and interpret insights easily.

7. DASHBOARD DESIGN OVERVIEW:

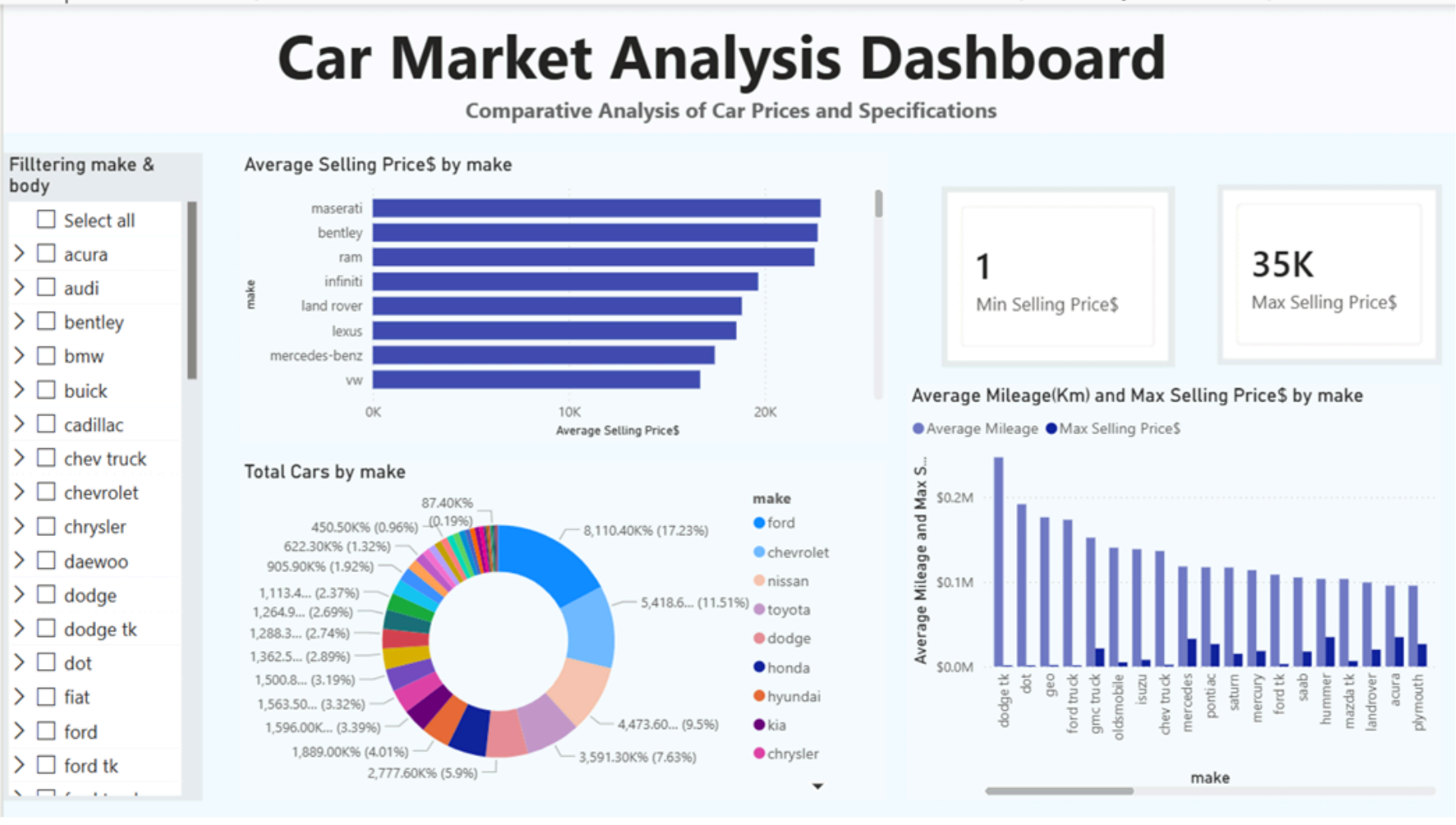


Figure 1: Car Market Analysis Dashboard – Pricing and Market Distribution Page

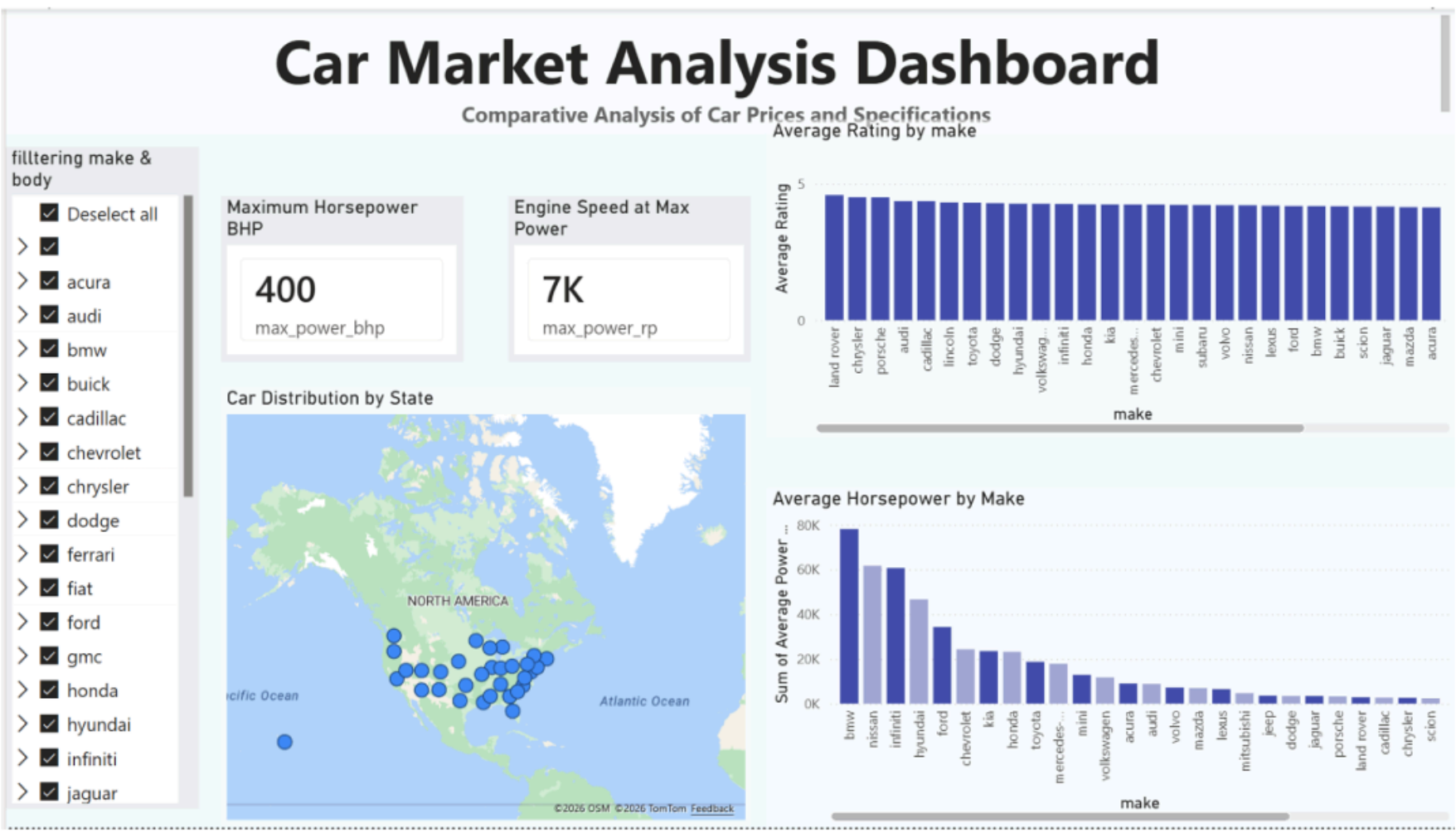


Figure 2: Car Market Analysis Dashboard – Specifications and Performance Page



8. IMPLEMENTATION:

8.1 Custom Measures:

A. Measures in car_prices_BI Table:

1. **Average Mileage:** Calculates the average mileage of cars and Helps analyze vehicle usage and driving distance.
2. **Average Selling Price:** Calculates the average selling price Used to compare prices between car brands.
3. **Max Selling Price:** Identifies the highest selling price Highlights premium-priced vehicles.
4. **Min Selling Price:** Identifies the lowest selling price Helps analyze affordable vehicle options.
5. **Price Difference:** Calculates differences between car prices Supports comparative price analysis.
6. **Total Cars:** Counts the total number of cars Provides an overview of dataset size.

B. Measures in car_Specifications_BI Table

1. **Average Power per Make:** Calculates average horsepower for each brand
Supports performance comparison between manufacturers.
 2. **Average Rating:** Calculates the average car rating. Helps evaluate customer satisfaction across brands.
- 



8. IMPLEMENTATION:

8.2 Dashboard Layout:

The dashboard title is centered to maintain focus on the main analysis topic. Interactive features such as slicers are placed on the left side, allowing users to filter data and explore specific car brands and categories easily. KPI cards are positioned clearly to provide quick insights into important metrics such as selling prices and car performance. Visualizations are arranged centrally to support fast and effective analysis while maintaining a clear and organized dashboard layout.



8. IMPLEMENTATION:

8.3 Visuals

The dashboard visuals were carefully selected to represent the data effectively, identify patterns, and generate meaningful insights. Different visualization types such as bar charts, KPI cards, donut charts, and maps were used to support comparative analysis and improve data interpretation.

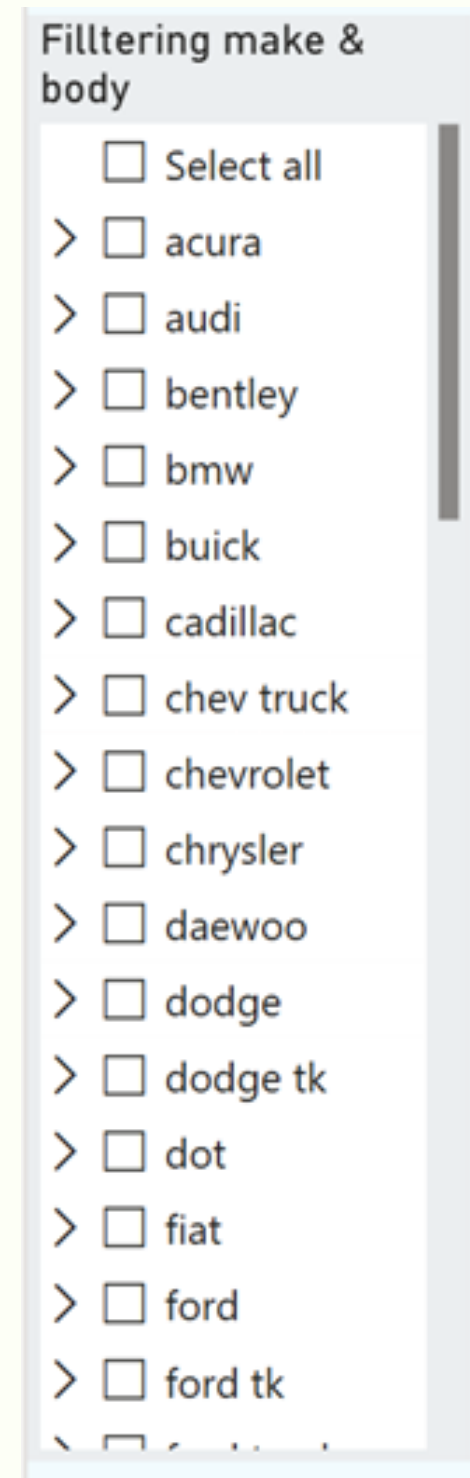


Figure 3:Slicer

8.3.1 Interactive Slicer

Interactive slicers were added to the dashboard to allow users to filter and explore the data dynamically based on car make and body type.

This feature improves dashboard and helps users compare vehicle categories and analyze specific brands more effectively.

8. IMPLEMENTATION:

8.3.2 Map

A map visualization was used to display the geographical distribution of cars across different states and regions.

The map helps users identify vehicle concentration in various locations and provides quick geographical insights into regional distribution patterns within the dataset.

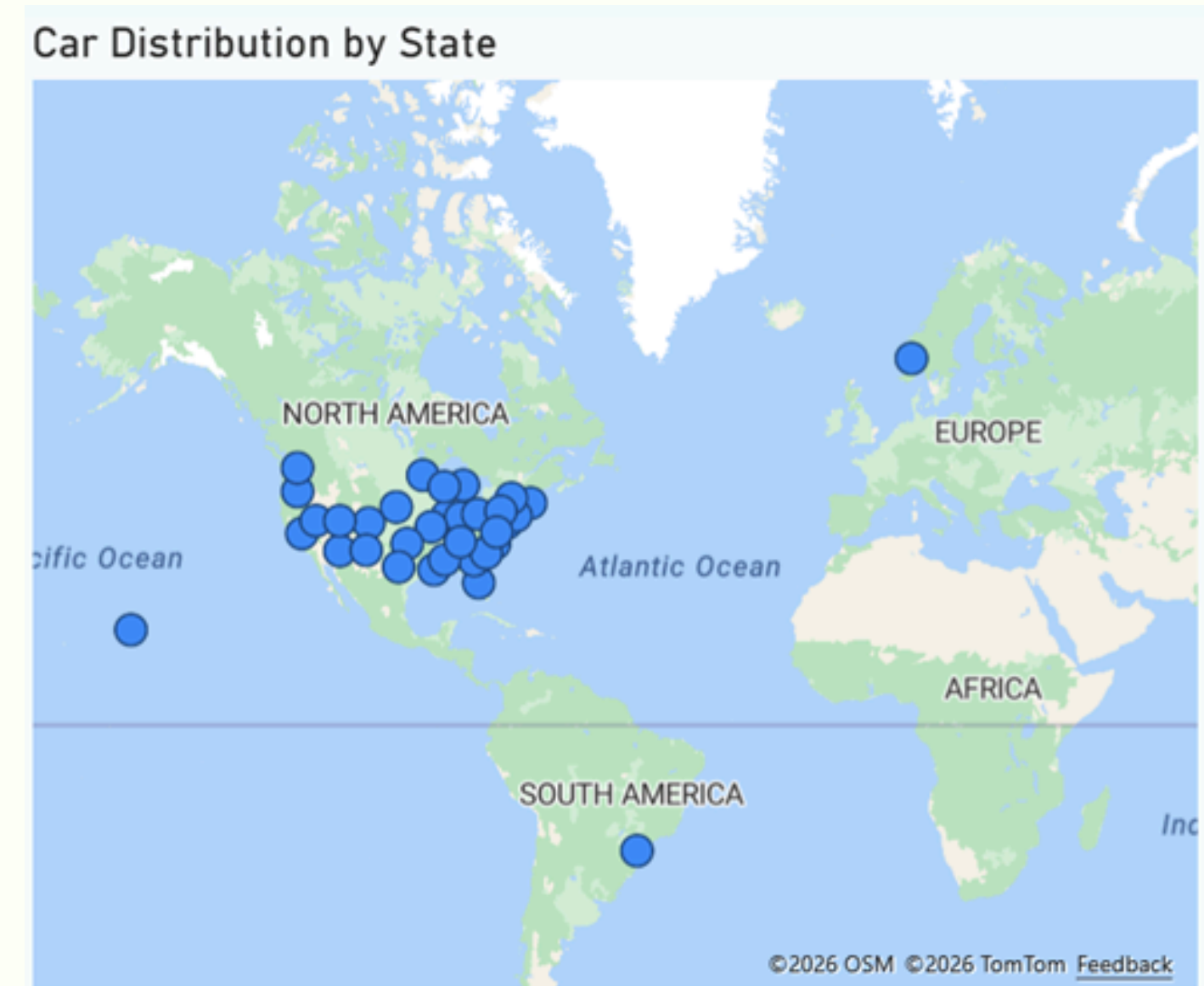


Figure 4:Map

8. IMPLEMENTATION:

8.3.3 Clustered Bar Chart

A clustered bar chart was used to compare the average selling prices of different car brands.

This visualization helps identify brands with higher and lower market values.

The analysis showed that brands such as Maserati, Bentley, and Ram recorded higher average selling prices compared to other manufacturers in the dataset.

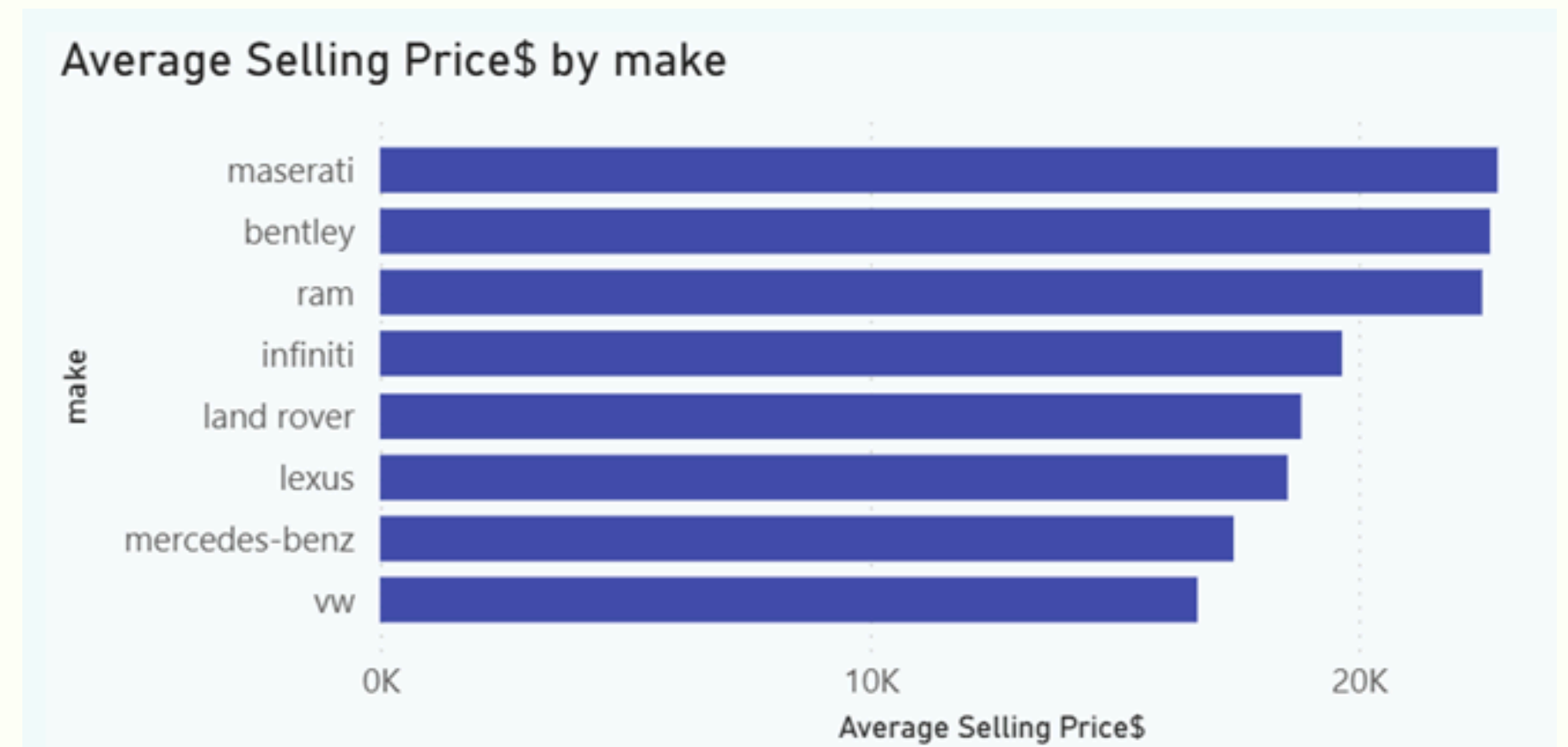


Figure 5: Clustered bar chart

8. IMPLEMENTATION:

8.3.4 Clustered Column Charts

Clustered column charts were used to compare performance and pricing metrics across different car brands.

The Average Horsepower chart highlights variations in engine performance, where brands such as BMW, Nissan, and Infiniti recorded higher horsepower values.

The Average Rating chart compares customer ratings between manufacturers and shows that most brands received relatively high and similar ratings.

In addition, the **Average Mileage and Max Selling Price chart** was used to analyze the relationship between vehicle usage and pricing, providing insights into market value differences across manufacturers.

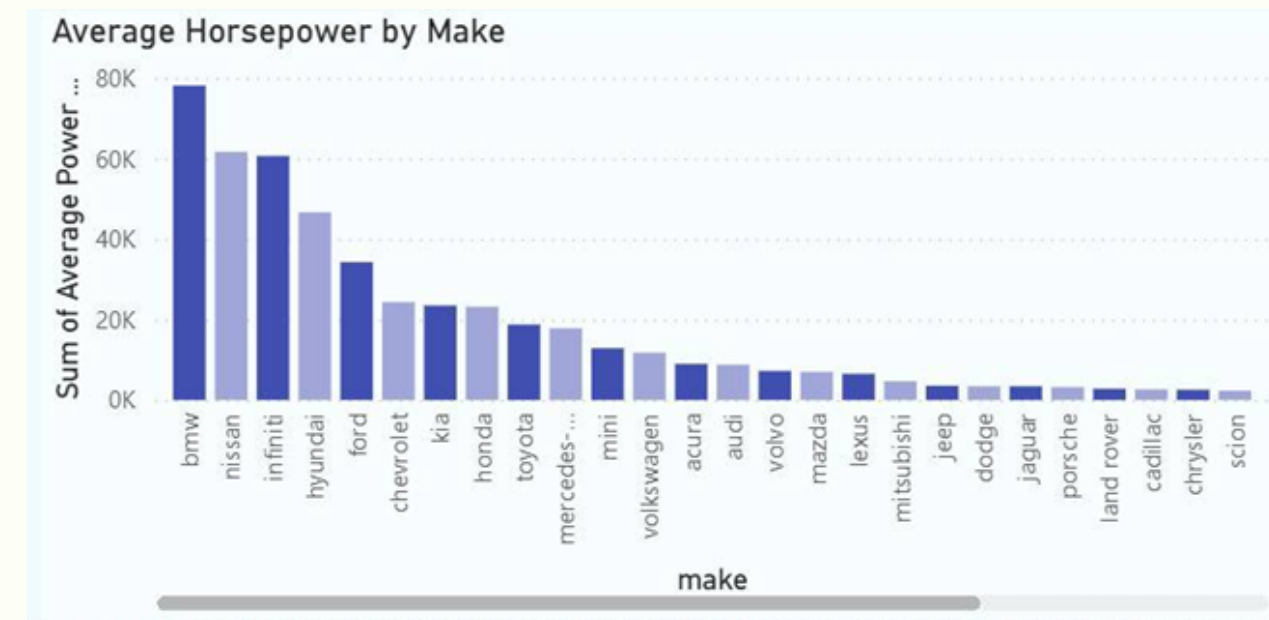


Figure 6 :Clustered Column Chart – Average Horsepower by Make



Figure 7: Clustered Column Chart – Average Mileage and Max Selling Price by Make

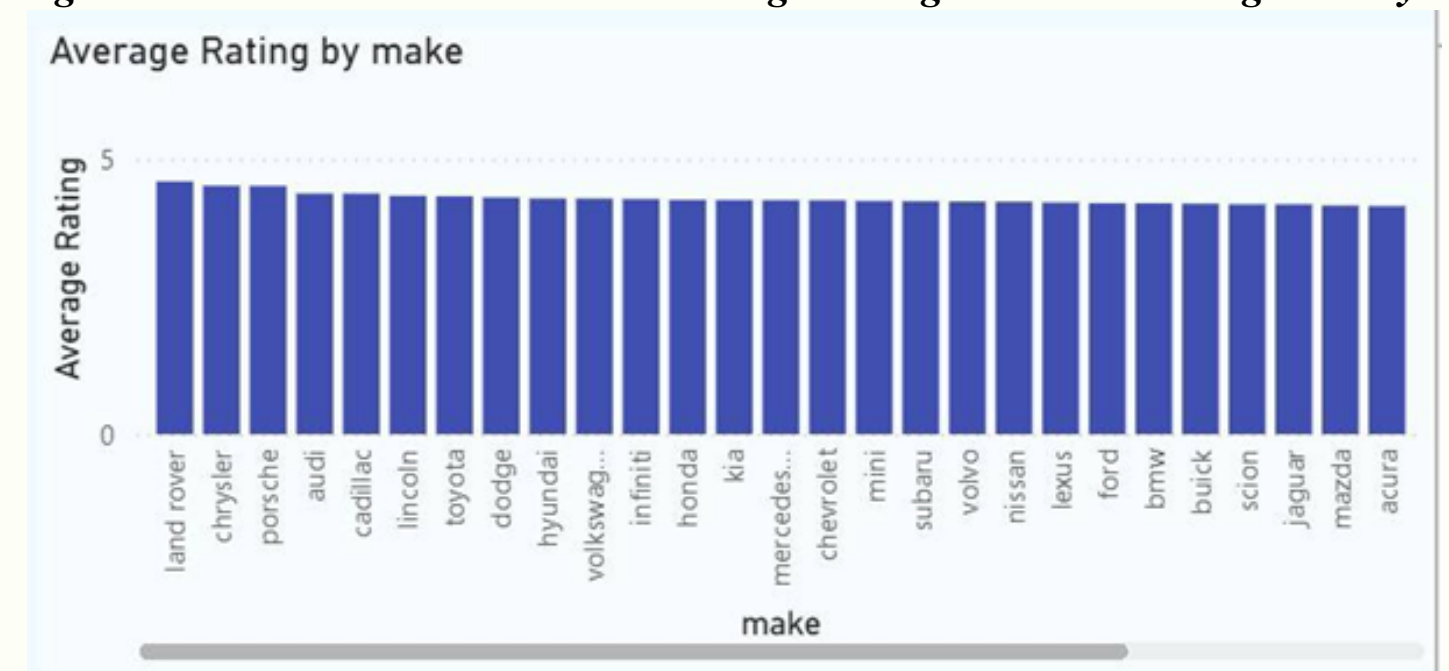


Figure 8:Clustered Column Chart – Average Rating by Make



8. IMPLEMENTATION:

8.3.5 KPI Cards:

KPI cards were used to display important summary metrics and provide quick insights into car pricing and vehicle performance.

The first KPI cards present the minimum and maximum selling prices, helping users identify the vehicle price range quickly.

Additional KPI cards display the maximum horsepower (BHP) and engine speed at maximum power (RPM), allowing efficient analysis of vehicle performance and engine capability.

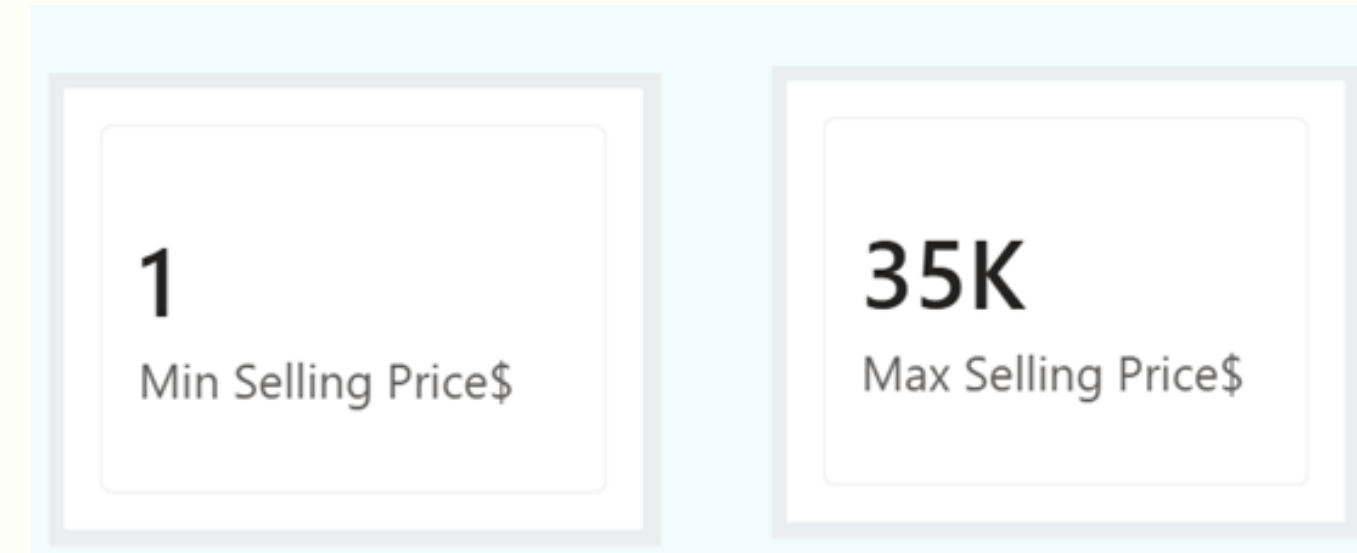


Figure 9: KPI Cards – Minimum and Maximum Selling Prices

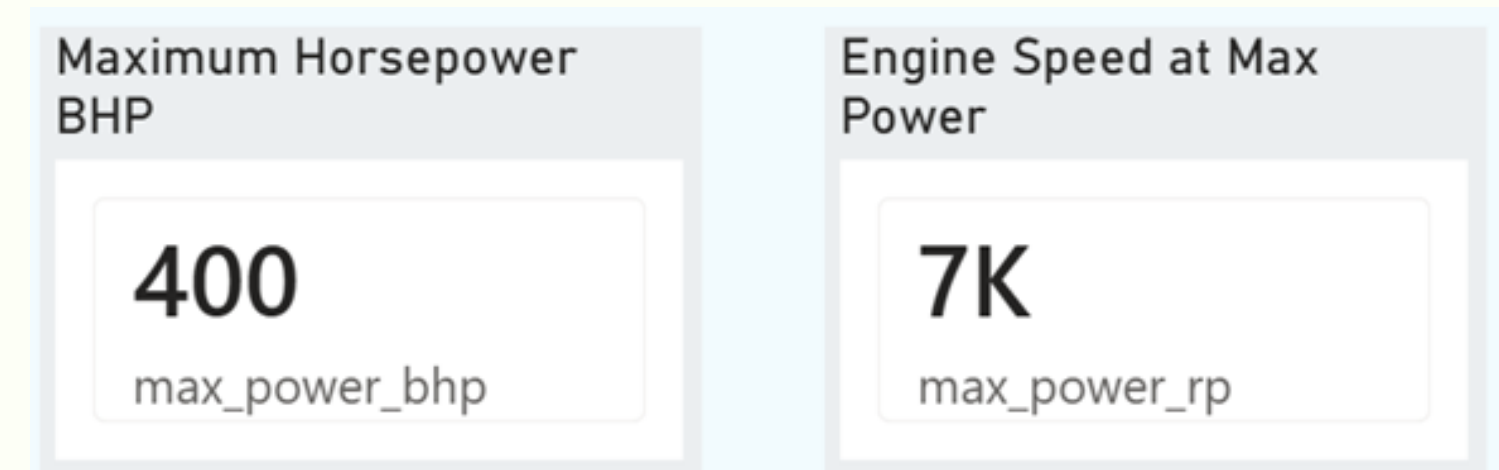


Figure 10: KPI Cards – Maximum Horsepower and Engine Speed at Maximum Power

8. IMPLEMENTATION:

8.3.6 Donut Chart:

A donut chart was used to display the distribution of cars by make and show the proportion of each car brand within the dataset.

The chart simplifies comparisons between manufacturers by representing each brand as a percentage of the total number of cars.

The analysis showed that brands such as Ford, Chevrolet, and Nissan represented a larger share of the dataset compared to other manufacturers.

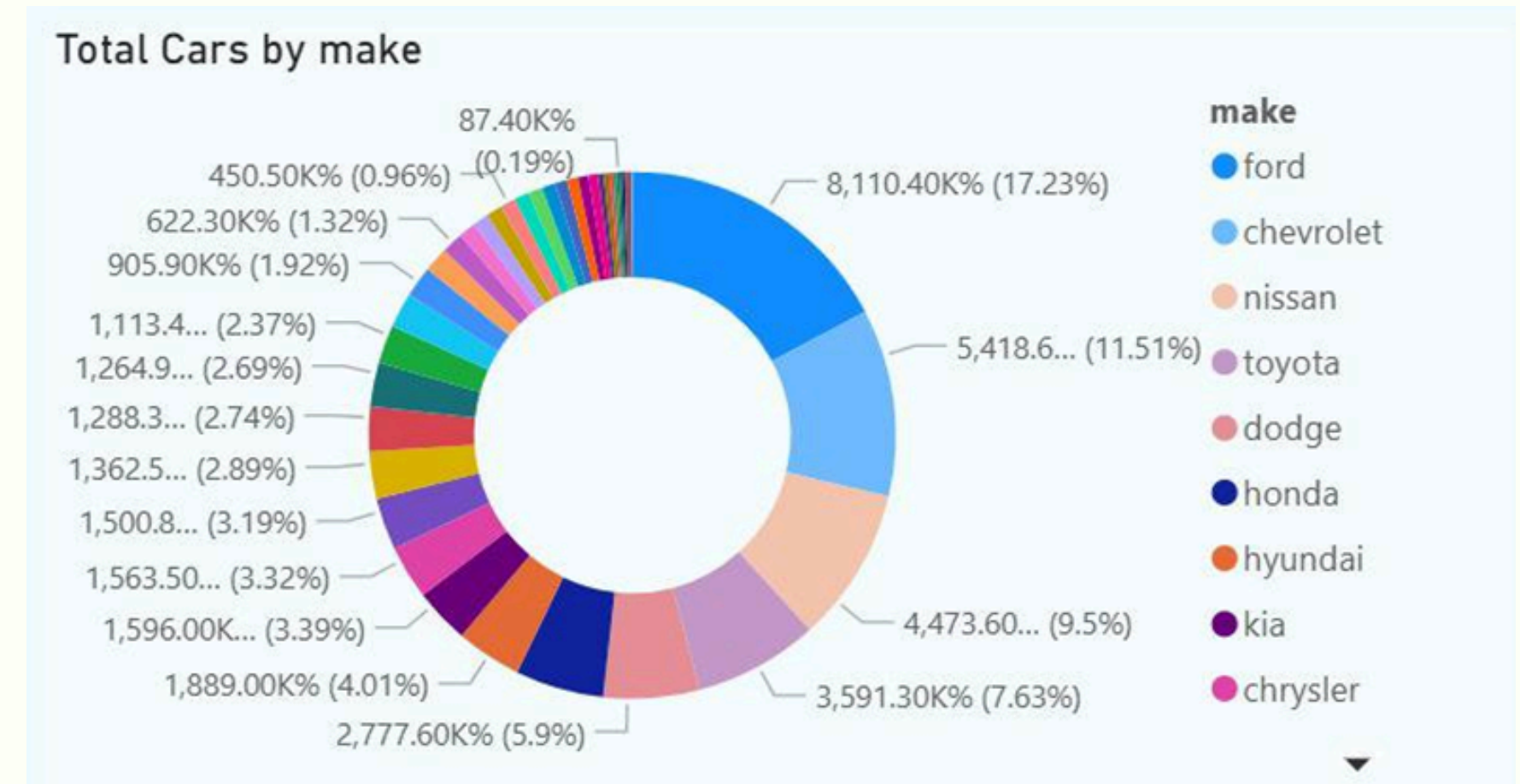


Figure 11: Donut chart



9. POWER BI FEATURES USED:

1. Filters and Slicers: Enabled dynamic filtering by car make and body type, allowing users to focus on specific segments.
2. KPI Cards: Used to display important metrics such as maximum horsepower (BHP), engine speed (RPM), and selling prices.
3. Charts and Visuals: Implemented clustered column charts and donut charts to compare average ratings, horsepower, selling prices, and total car counts.
4. Map Visualization: Applied to display car distribution by state, adding a geographical dimension to the analysis.
5. Data Modeling: Relationships were established between tables to ensure accurate filtering and consistent data aggregation.



10. PROBLEMS FACED:

During the development of the dashboard, several challenges were encountered.

One of the main issues was establishing correct relationships between tables because the Car_Key column contained duplicate values, which caused many-to-many relationship problems.

To solve this issue, an additional table with unique Car_Key values was created to support proper one-to-many relationships.

The project also faced data cleaning challenges, including duplicate values, missing data, and inconsistent formatting in columns such as make and model. In addition, configuring slicers and ensuring that filters interacted correctly with all visuals required multiple adjustments.

Several modifications were also needed to improve dashboard layout, alignment, colors, and overall readability. Finally, some numerical values initially appeared in incorrect formats, such as percentages instead of currency values, which required additional formatting corrections.



11. LIMITATIONS:

Despite the effectiveness of the dashboard, several limitations were identified during the project. Some car brands contained a limited number of records compared to others, which may affect the accuracy of comparisons and overall analysis.

In addition, the tables did not include important features such as fuel efficiency, accident history, or maintenance records, which could provide deeper insights into vehicle performance and pricing.

Some visualizations were also limited by the available data structure and Power BI visual settings. Furthermore, the analysis depends on the quality and completeness of the original tables, meaning that missing or inconsistent values may influence the final insights.





12. FUTURE WORK:

For future work, the dashboard could be enhanced by integrating additional tables related to fuel efficiency, maintenance history, accident records, and customer reviews to provide deeper market insights.

Advanced analytical techniques such as predictive analytics and machine learning models could also be applied to predict car prices and analyze market trends more accurately.

In addition, more interactive features and real-time data updates could be added to improve user experience and support better decision-making.





13. CONCLUSION:

In this project, an interactive Car Market Analysis Dashboard was developed using Power BI to analyze car prices, specifications, and vehicle performance across different manufacturers. The dashboard transformed raw data into meaningful visual insights using KPI cards, charts, maps, slicers, and data modeling techniques.

The analysis provided valuable insights into selling prices, horsepower, mileage, ratings, and geographical distribution of vehicles. Overall, the project demonstrates how Power BI can support data analysis and improve decision-making through effective visualization and interactive dashboard design.





15. REFERENCES:

- [1] United Nations, "Goal 9: Industry, Innovation and Infrastructure," Sustainable Development Goals, 2023. [Online]. Available: <https://sdgs.un.org/goals/goal9>.
- [2] B. Tunguz, "Used Car Auction Prices Dataset," Kaggle, 2022. [Online]. Available: https://www.kaggle.com/datasets/tunguz/used-car-auction-prices?select=car_prices.csv.
- [3] T. Tr1gg3rtrash, "Cars 2022 Dataset," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/tr1gg3rtrash/cars-2022-dataset>.

THANK
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ANY QUESTIONS?

